Information Technology Department

Final Examination
Fall, AY 2022/2023
ITSE202 – Introduction to Software Engineering

Date: Sun, Jan 8, 2023 **Time:** 9:00 AM – 11:00 AM

Max Marks	40	Time Allocation	2 Hours

Student ID		Student Name	
Section		Level	Diploma Year II

Instructions to the Students

1. Answer the questions in the space provided by using **blue or black pen** only. You will obtain a zero (0) if you use a pencil for all questions.
2. Use of mobile phones and any electronic gadgets inside the examination hall are not allowed.
3. ***Copying, Cheating, and any kind of malpractices on the examination are strictly prohibited.*** Action will be taken as per the "Article 80, Bylaws of Colleges of Technology".
4. Students should bring their own pencils, pens, erasers, rulers, calculators, and any other tools required for the exam.
5. The exam has a total of **16 non optional questions** in all.
6. **Only Scientific Calculators are allowed.**

Section	Allotted Marks	Obtained Marks
Section A	**8**	
Section B	**15**	
Section C	**17**	
Total Marks	**40**	

Marked By	Checked/Re-marked by

Section A: Multiple Choice Questions (8 X 1 = 8 marks)

1)	Which one of the following is the correct data flows in data flow diagrams.	
	a) entity to process	b) entity to entity
	c) data store to entity	d) data store to data store
2)	The triple constraints on a project are:	
	a) Time, Cost, Profit	b) Resources, Time, Cost
	c) Time, Cost, Quality	d) Resources, time, Profit
3)	Which one of the following notations is used for "entity" in Yourdon & Coad's DFD notations?	
	a) Circle	b) Square
	c) Open-ended box	d) Arrow
4)	Software testing includes	
	a) Verification only	b) Validation only
	c) Risk Analysis	d) Verification & Validation
5)	"There will be an extension for the project schedule for 20 days because of the delay from the Supplier from muscat." The above risk falls under which one of the following categories?	
	a) Schedule Risk	b) Resource Risk
	c) Scope Risk	d) Programmatic Rick
6)	Identify the invalid statement: Software engineers shall	
	a) act with the public interest	b) supportive of their colleagues
	c) develop s/w only meet the SRS	d) participate in lifelong learning
7)	"It is an informal debate held among various stakeholders and all their inputs are recorded for further requirements analysis. The above scenario is an example of _____ elicitation technique.	
	a) Brainstorming	b) Survey
	c) Interview	d) Task Analysis
8)	Which type of cohesion is most used or wanted?	
	a) Logical	b) Procedural
	c) Coincidental	d) Functional

Section B: Short Answer Questions (15 marks)

9) Based on the given logical scenario, identify the suitable process model. [3 marks]

Scenario: Staqo World LLC is a software company that is developing the Course Management System for UTAS. As the system is big and complex. the developers need to identify more than one alternative solution. The development team decided to prepare prototypes for the software to minimize the risk and system errors.

Process Model:

Reasons:

10) List any two differences between `White Box Testing` and `Black Box Testing`. [2 marks]

	White Box Testing	Black Box Testing
1)		
2)		

11) a) Assume that a Java program needs 65 LOC per one function point. Calculate the Function Points for an average project with `10 inputs, 2 outputs, 3 inquiries, 4 master files, and 5 interfaces`. Hence, estimate the size of the project. [2 marks]

(Note: Write all the steps in your calculations.)

Levels of complexity			
Component	**Simple**	**Average**	**Complex**
Input	3	4	6
Output	4	5	7
Inquiry	3	4	6
Master File	7	10	15
Interface	5	7	10

Total FP =

Estimated Size=

b) A construction company is planning to develop a new computer program to keep track of the materials in their store. An initial study has determined that the size of the program will be roughly 95,000 delivered source instructions. This project is a good example of an organic mode software project. Using the basic COCOMO equations calculate the following estimations. [3 marks]

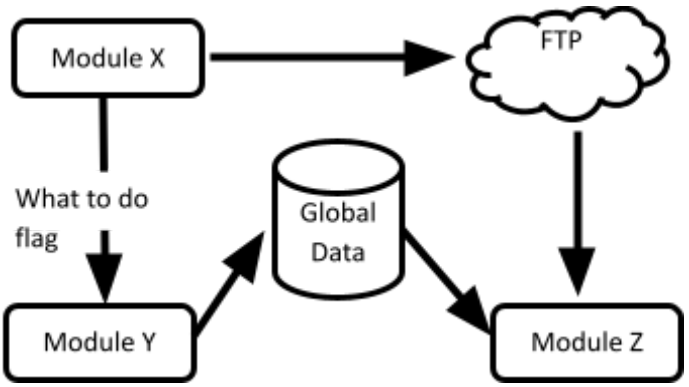
Software Project	a_b	b_b	c_b	d_b
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Effort (E) =

Time to develop (TDEV) =

Average Staffing (SS) =

12) Analyze the given diagram and identify the type of couplings between the modules. [3 marks]



Coupling between	Coupling Name
Module X □ Module Y	
Module Y □ Module Z	
Module X □ Module Z	

13) Identify any two (2) responsibilities of a project manager. [2 marks]

Section C: Scenario-Based Questions (17 marks)

14) Read the below scenario carefully and draw the `level-1` DFD for the given scenario. Identify and draw all related external entities, processes, data flows, and data stores. [5 marks]

Scenario: Moodle page allows the teacher to login to the system with username and password. Upon successful login, the teacher will save the assignment question to the system. The HOS will approve the question. The system will save the approved details like the approved date. The student will submit the answer to the assignment question. The student's answer, date and time will be recorded by the system. The registration department can manage (add/delete/update) teachers on Moodle page.

15) **Scenario:** You are assigned to develop the payment module for the VOX cinema ticketing system. The module allows an applicant to enter Billing information like name, phone number, address, and the Payment Details like card type (card type can be Visa or MasterCard), card number, expiry month and expiry year, a three-digit secure CVV, and amount. When the user pays the amount, the payment details will be validated by the bank and will automatically send an OTP (One-Time Password) to the applicant. The user needs to enter the OTP to Complete Payment.

a) Draw the `Graphical User Interface` for the given Payment Module scenario. [3 marks]

(Note: Provide at least six (6) suitable menu items in your interface.)

b) Create a `Test Case` (at least 3 possible) for the given Payment Module scenario. [3 marks]

CaseID	Action	Expected Result	Data	Actual Result	Test Case Result

16) Convert the Al Hoota Cave ticketing system scenario into Decision Table and Structured English.

Scenario: If the visitor is Omani and the age is more than 60, the ticket price is 2 OMR. If the visitor is Omani or the age is more than 60 then the ticket price is 3 OMR. Otherwise, ticket price is 5 OMR.

Decision Table: [3 marks]

Structured English: [3 marks]